EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 02

Vectors

Components and Resultant

By - Excellence Programme

29 September 2026

Topics to be covered

01 Types of vectors, part 2

02 Multiplying a vector by a scalar

03 How to read east of south

04 Rectangular components and magnitude

05 Triangle law and the resultant formula

06 Subtraction of vectors

07 Addition by components

08 Dot product, angle, and unit vector

Types of vector — equality and lines

Equal: same magnitude, same direction, same kind of quantity.

Parallel: same direction. Magnitudes may differ. Every equal pair is parallel. A parallel pair need not be equal.

Negative: same magnitude, exactly opposite direction.

Anti-parallel: opposite direction, magnitudes may differ. Every negative pair is anti-parallel. An anti-parallel pair need not be a negative pair.

Types of vector — size and plane

Unit vector: magnitude 1, no unit, direction only.

Zero vector: magnitude 0. Example: velocity of a car at rest.

Collinear: all lie on the same straight line.

Coplanar: all lie in the same plane.

Orthogonal: angle between them is 90°. Magnitudes may differ.

Concurrent: arrows act at the same point, even if magnitudes and directions differ.

Axial: direction is along the axis of rotation. Examples: torque, angular velocity, angular acceleration, angular momentum.

Direction does not make every quantity a vector

A: a quantity with no direction must be a scalar. **B: a quantity with a direction must be a vector.**

Solution: Direction does not make every quantity a vector

Step 1: A is true. A scalar has no direction.

Step 2: B is false. Current has a direction but does not obey the triangle law.

A true, B false.

Rough work / extra work

Types of vectors — lock lines

Type	Magnitude	Angle between them
Equal, same kind	same	0°
Parallel	may differ	0°
Opposite (negative)	same	180°
Anti-parallel	may differ	180°
Orthogonal	may differ	90°

Unit, null, collinear, coplanar

Unit vector: magnitude 1, no unit. $\hat{A} = A / |A|$.

Null vector: magnitude 0. Direction is not defined. Example: two equal opposite forces, net force 0.

Collinear: along one straight line, so the angle is 0° or 180°.

Coplanar: every arrow lies in the same plane.

Concurrent: the arrows pass through one common point.

Axial: along the axis of rotation. Examples: angular velocity, angular momentum, torque. Moment of inertia is not a vector.

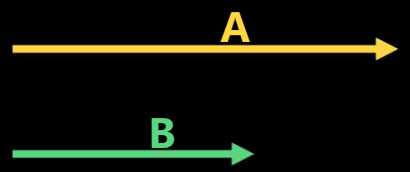
Equal, parallel, opposite, anti-parallel

Do not remember these by words only. Compare three things: kind, length and direction.



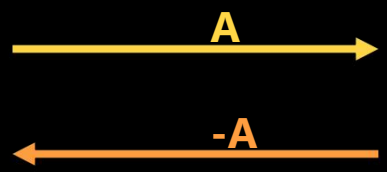
Equal vectors

same kind + same length + same direction



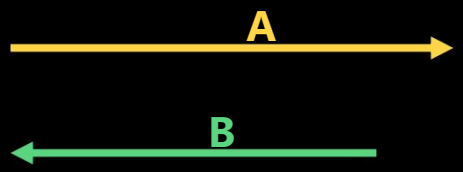
Parallel

same direction, length may differ



Opposite / negative

same length, exactly 180° direction



Anti-parallel

opposite direction, length may differ

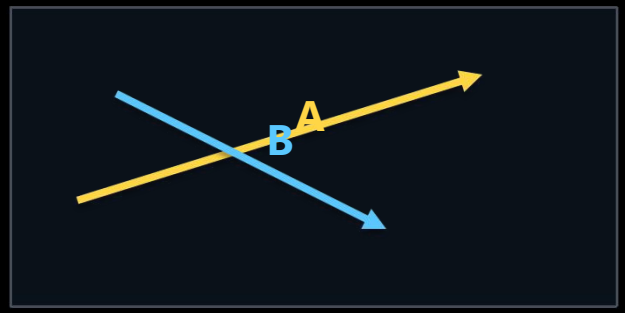
Exam trap: equal is stronger than parallel. Opposite is stronger than anti-parallel.

Line, plane, point and axis

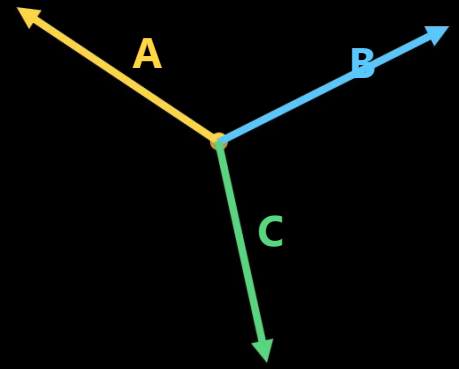
These names tell you where the arrows live geometrically.



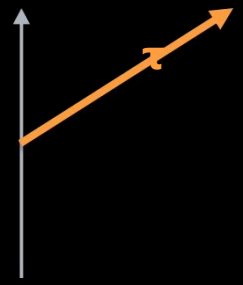
Collinear



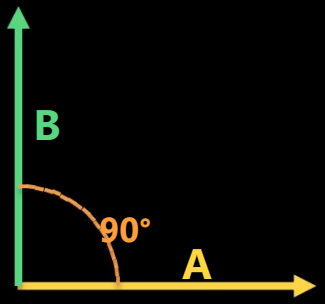
Coplanar



Concurrent



Axial vector: along rotation axis



Orthogonal

Rough work / extra work

Which statements are true?

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1. Every parallel pair is equal. 2. Every equal pair is parallel. 3. Every anti-parallel pair is opposite. 4. Every opposite pair is anti-parallel.

Solution: Which statements are true?

Step 1: 1 is false. Parallel vectors may have different lengths.

Step 2: 2 is true. Equal vectors have the same direction.

Step 3: 3 is false. Anti-parallel lengths may differ.

Step 4: 4 is true. Opposite means same length and 180°, so they are anti-parallel.

Same number and same direction

F = 20 N east and v = 20 m/s east. Are F and v equal vectors?

Solution: Same number and same direction

Step 1: Equal vectors must be the same kind of quantity

Step 2: Force and velocity are different quantities, so they cannot be equal

Not equal. Two 20 N forces both east are equal.

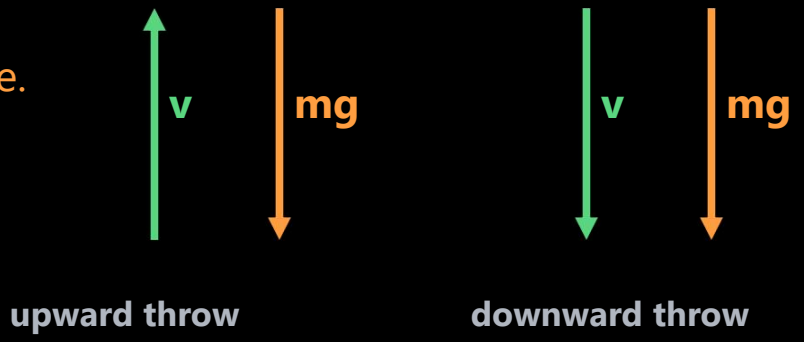
Thrown up and thrown down

Weight mg is always downward.

Object projected upward: velocity up, weight down. They are anti-parallel.

Object thrown downward: velocity and weight are both down. They are parallel.

Parallel means the same direction. It does not demand equal magnitude.



Rough work / extra work

Multiplying by a scalar

Positive scalar: magnitude changes, direction stays.

Negative scalar: magnitude changes, direction reverses.

What a scalar does to an arrow

A scalar does not create a new direction unless the scalar is negative.



positive scalar: same direction



negative scalar: reversed direction

Board line

kA keeps A's line. k > 0 keeps direction. k < 0 reverses direction.

3F and minus 2F

F = 10 N î. Find 3F, F/2 and -2F.

Solution: 3F and minus 2F

Step 1: 3F = 30 N î. Same direction, longer.

Step 2: F/2 = 5 N î. Same direction, shorter.

Step 3: -2F = -20 N î. Opposite direction.

The minus sign flips the arrow. It does not change the kind of quantity.

$\mathbf{A = 30\,\hat{i} + 20\,\hat{j}}$

Find 2A, A/2 and -5A.

Solution: A = 30 î + 20 ĵ

Step 1: 2A = 60 î + 40 ĵ

Step 2: A/2 = 15 î + 10 ĵ

Step 3: -5A = -150 î - 100 ĵ

Multiply every component. A minus outside flips both.

Rough work / extra work

$\mathbf{A} = 30\ \mathrm{m}\ \hat{\imath}$

Find 3A, A/3, -5A, the magnitude of A, and the unit vector.

Solution: A = 30 m î

Step 1: 3A = 90 m î and A/3 = 10 m î

Step 2: -5A = -150 m î

Step 3: |A| = 30 m. The unit vector is î, with no unit.

Unit vector = vector / magnitude.

How to read a compass phrase

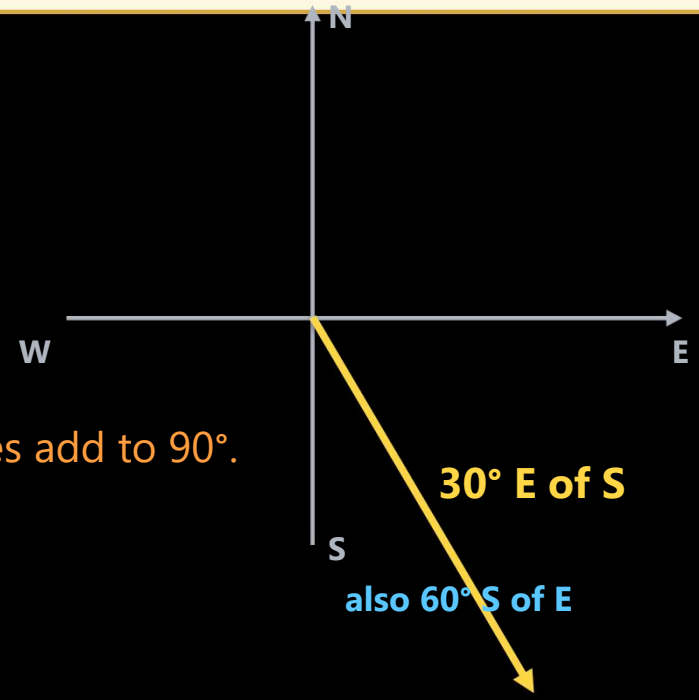
The last word is the reference line. The first word is the side you turn toward.

30° east of south: start facing south, turn 30° toward east.

60° south of east: start facing east, turn 60° toward south.

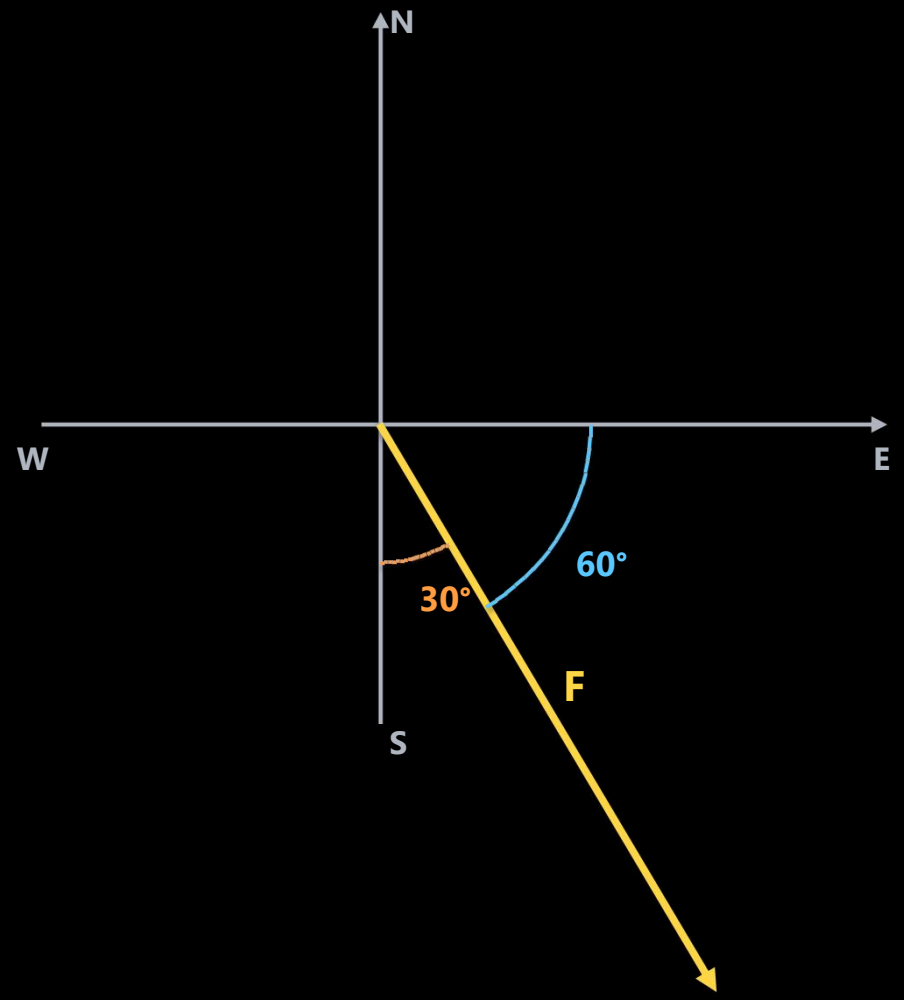
Those two phrases describe the same arrow.

East of south is not the same wording as south of east, even when the angles add to 90°.



Compass phrases: one ray can have two names

Start from the last word. Turn toward the first word. The same slanted ray can be described from either nearby axis.



30° east of south

Reference = south. Turn 30° toward east.

60° south of east

Reference = east. Turn 60° toward south.

Both names point to the same ray.

Rough work / extra work

Angle between the equal vectors changes when one is reversed

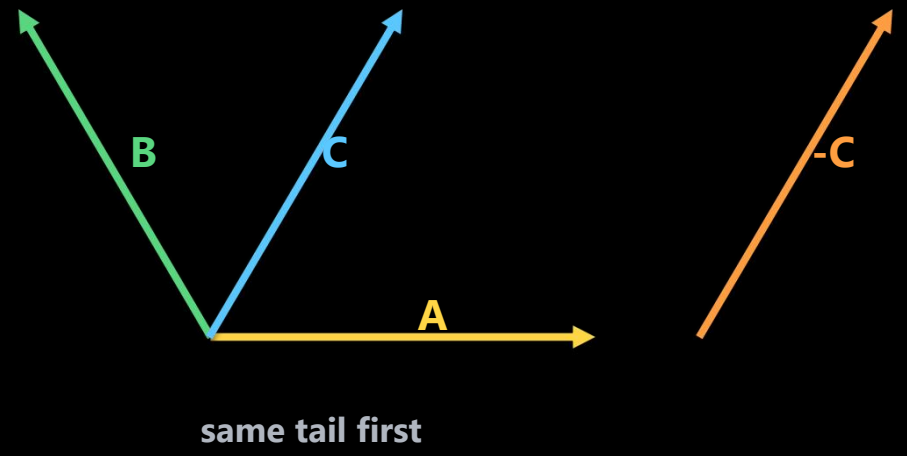
In an equilateral layout, angle between A and B is 120°. Find the angle between A and -B, and between A and -C if angle A with C is 60°.

Solution: Angle between the equal vectors changes when one is reversed

Step 1: Reversing one arrow changes 120° into 180 - 120 = 60

Step 2: If angle between A and C is 60, angle between A and -C is 120

Flip one arrow, and the angle becomes its supplement.



same tail first

Angle 30°, then multiply

Angle between A and B is 30°. Find the angles between A and -B, between -A and -B, and between 2A and -3B.

Solution: Angle 30°, then multiply

Step 1: A positive multiplier does not change direction, so 2A stays with A

Step 2: A minus flips the arrow, so angle with -B is 150°

Step 3: Two minuses bring both arrows back, so -A and -B stay at 30°

2A and -3B are 150° apart. 2A and B stay at 30°.

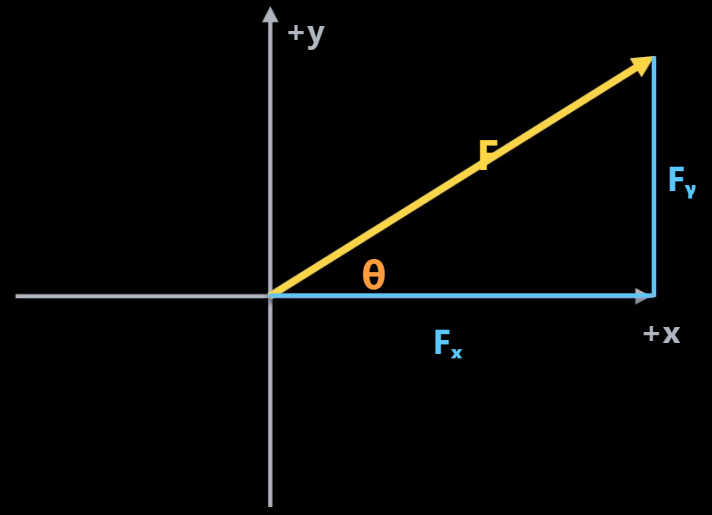
Rectangular components

If θ is measured from the +x axis: $F_x = F \cos \theta$, $F_y = F \sin \theta$

$F = \sqrt{(F_x^2 + F_y^2)}$ and $\tan \theta = F_y / F_x$

The sign of a component is the sign of that axis. Left of the y-axis means a negative x-component. Below the x-axis means a negative y-component.

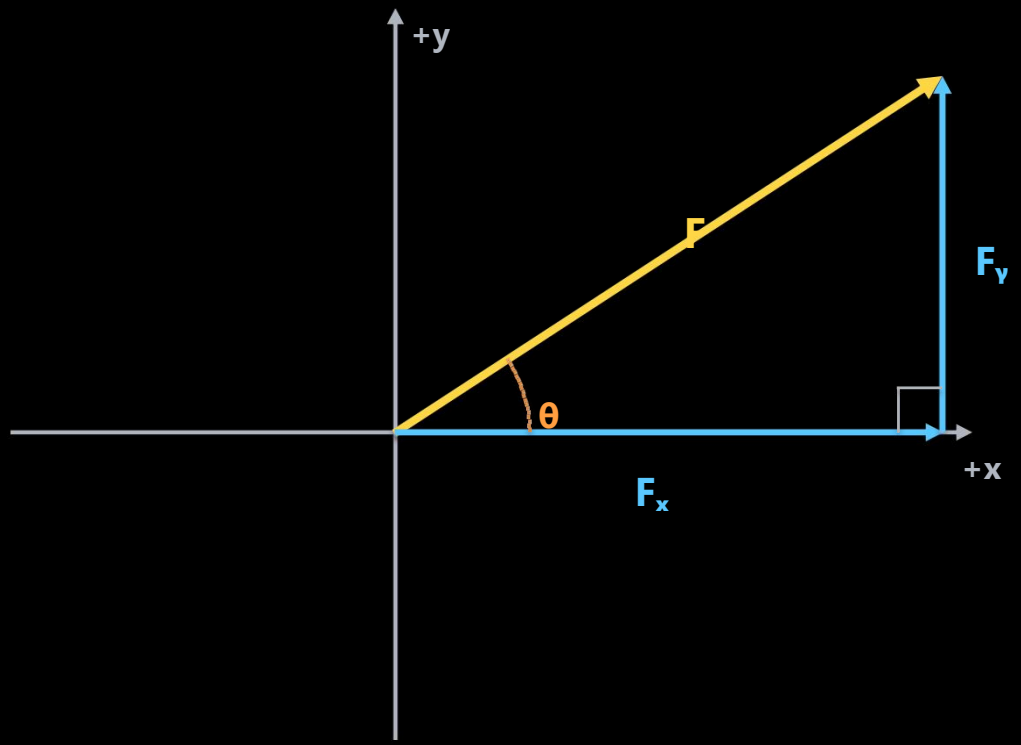
Cos goes with the axis you measured the angle from. Do not swap sin and cos.



Rough work / extra work

Resolving a vector means making a right triangle

A slanted vector can be replaced by two perpendicular vectors: one along x and one along y.



If θ is from x-axis

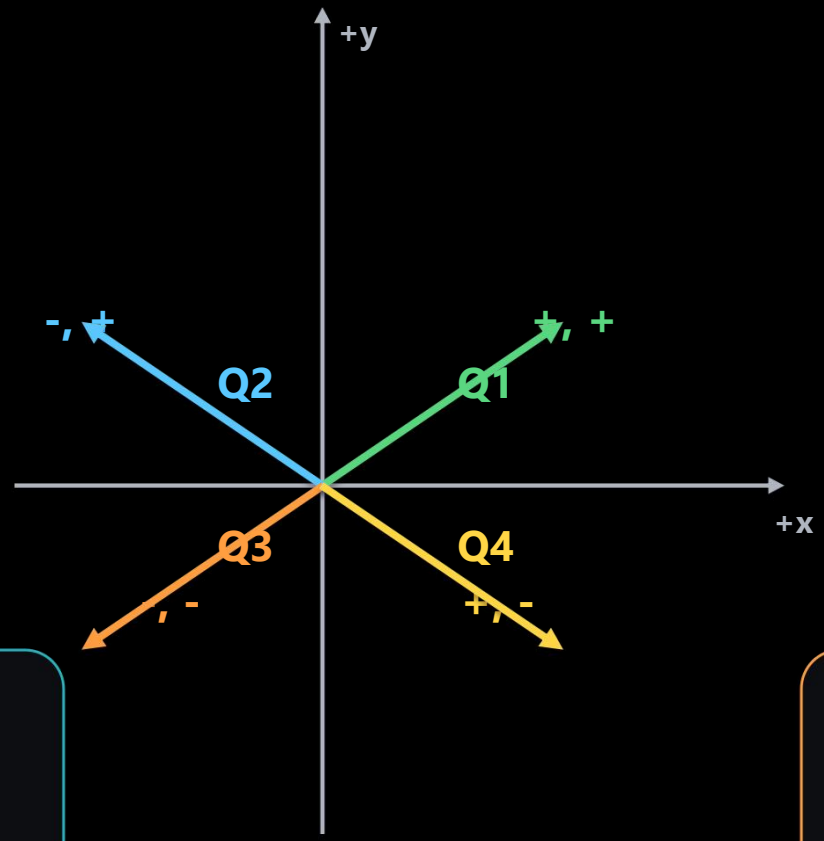
Adjacent side = $F_x = F \cos \theta$. Opposite side = $F_y = F \sin \theta$.

Sign comes later

First find size. Then attach + or - by quadrant.

Quadrant decides the signs

Magnitude is always positive. Components carry signs because axes have signs.



Memory line

Right means F_x positive. Left means F_x negative.
Up means F_y positive. Down means F_y negative.

Do not change sin/cos

Only the sign changes with quadrant.

37° and 53°

Angle	sin	cos
37°	3/5	4/5
53°	4/5	3/5
30°	1/2	√3/2
45°	1/√2	1/√2
60°	√3/2	1/2

10 N at 45°

A 10 N force is at 45° above the +x axis. Find both components.

Solution: 10 N at 45°

Step 1: $F_x = 10 \cos 45° = 10 / \sqrt{2} = 5\sqrt{2}$

Step 2: $F_y = 10 \sin 45° = 5\sqrt{2}$

Both components are $5\sqrt{2}$ N, and both are positive.

Rough work / extra work

10 N at 60°

A 10 N force is at 60° above the +x axis. Find F_x and F_y.

Solution: 10 N at 60°

Step 1: $F_x = 10 \cos 60° = 10 \times 1/2 = 5$

Step 2: $F_y = 10 \sin 60° = 10 \times \sqrt{3}/2 = 5\sqrt{3}$

$F_x = 5$ N, $F_y = 5\sqrt{3}$ N.

53° in the second quadrant

A 30 N force is in the second quadrant, 53° from the +y axis toward the -x axis. Find the components.

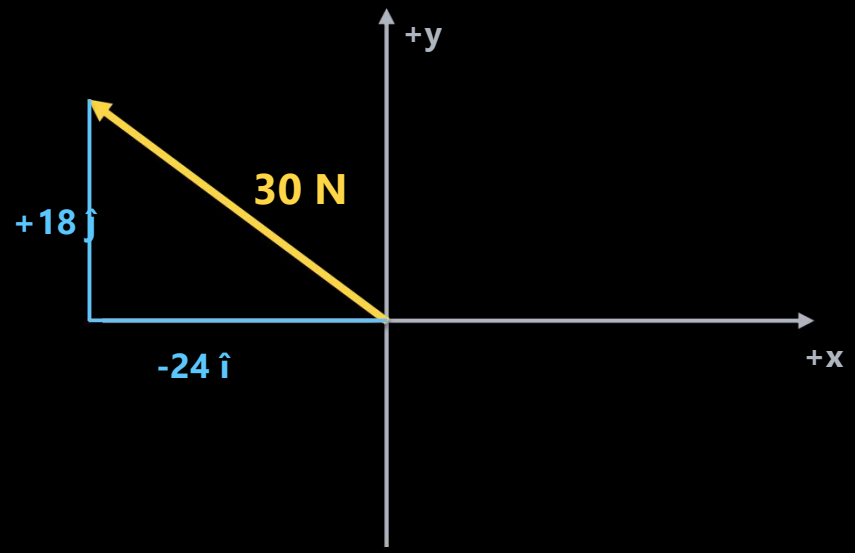
Solution: 53° in the second quadrant

Step 1: The angle from +y is 53°, so the y-component uses cos 53°

Step 2: $F_y = 30 \cos 53° = 30 \times 3/5 = 18$, so $+18\,\hat{j}$

Step 3: $F_x = 30 \sin 53° = 30 \times 4/5 = 24$, but toward -x

$F_x = -24\,\hat{i}$, $F_y = +18\,\hat{j}$.



50 N at 37° below +x

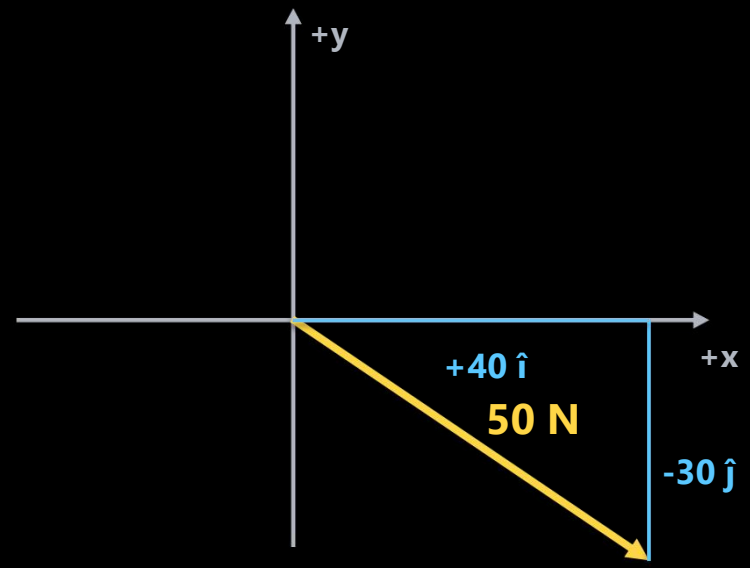
F = 50 N in the fourth quadrant, 37° below the +x axis. Find the components.

Solution: 50 N at 37° below +x

Step 1: $F_x = 50 \cos 37° = 50 \times 4/5 = 40$

Step 2: $F_y = -50 \sin 37° = -50 \times 3/5 = -30$

+40 î - 30 ĵ.



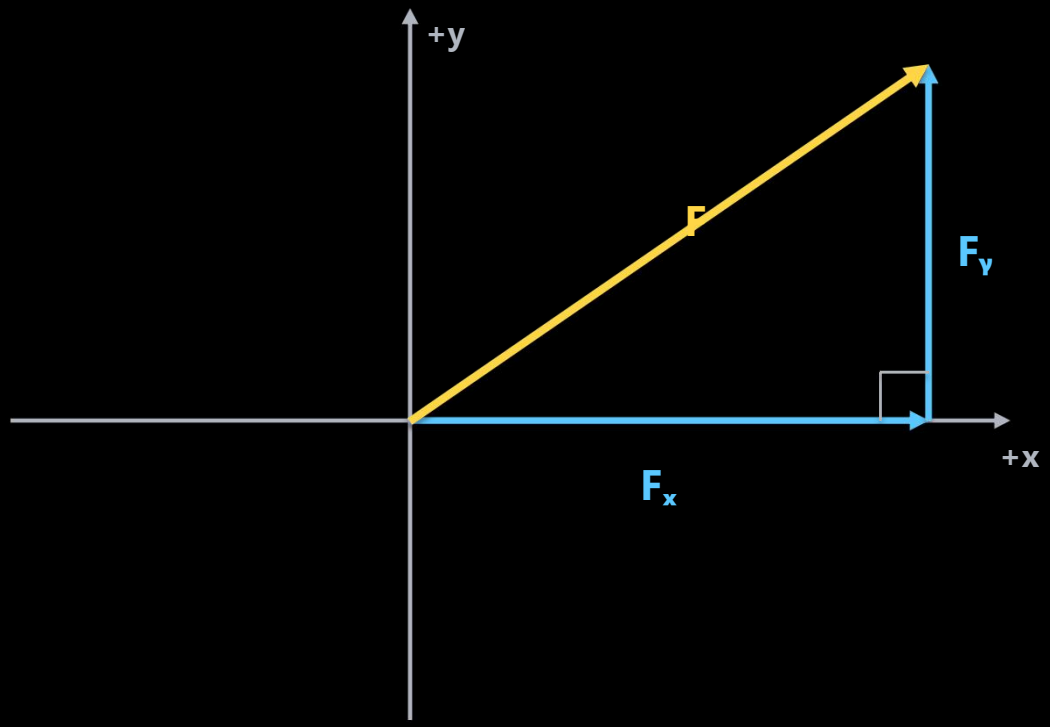
Rough work / extra work

Magnitude from components

$$|F| = \sqrt{(F_x^2 + F_y^2 + F_z^2)}$$

Magnitude is the length of the diagonal

Components form a right triangle. The vector is the diagonal, so Pythagoras gives its length.



$$F^2 = F_x^2 + F_y^2 \quad \text{so} \quad |F| = \sqrt{(F_x^2 + F_y^2)}$$

With z-component

In 3D, add one more perpendicular square: $|F| = \sqrt{(F_x^2 + F_y^2 + F_z^2)}$.

Magnitude of 30 î + 40 ĵ

Find the magnitude of F = 30 î + 40 ĵ.

Solution: Magnitude of 30 î + 40 ĵ

Step 1: $|F| = \sqrt{(30^2 + 40^2)} = \sqrt{(900 + 1600)}$

Step 2: $\sqrt{2500} = 50$

50

Rough work / extra work

Magnitude when a component is negative

Find $|2\,\hat{\imath} + 2\,\hat{\jmath}|$, $|10\,\hat{\imath} + 10\,\hat{\jmath}|$ and $|3\,\hat{\imath} - 4\,\hat{\jmath}|$.

Solution: Magnitude when a component is negative

Step 1: $\sqrt{(4 + 4)} = \sqrt{8} = 2\sqrt{2}$

Step 2: $\sqrt{(100 + 100)} = 10\sqrt{2}$

Step 3: $\sqrt{(9 + 16)} = \sqrt{25} = 5$. The minus does not enter the square.

$2\sqrt{2}$, $10\sqrt{2}$, and 5.

Three components

Find $|3\hat{i} + 4\hat{j} + 5\hat{k}|$ and $|6\hat{i} - 8\hat{j} + 10\hat{k}|$.

Solution: Three components

Step 1: $\sqrt{(9 + 16 + 25)} = \sqrt{50} = 5\sqrt{2}$

Step 2: $\sqrt{(36 + 64 + 100)} = \sqrt{200} = 10\sqrt{2}$

$5\sqrt{2}$ and $10\sqrt{2}$.

Triangle law of addition

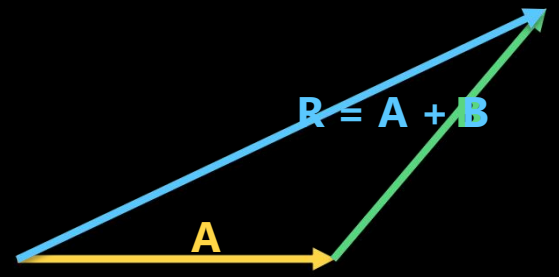
To find an angle, put tails together, or heads together.

To add, put the tail of the second arrow on the head of the first.

Resultant R runs from the first tail to the last head.

A + B = B + A. For three arrows the drawing order does not change R.

The resultant of two arrows lies between them, closer to the longer one.

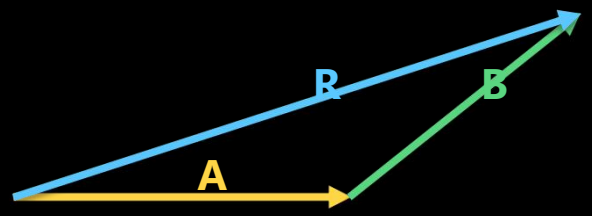


head-to-tail

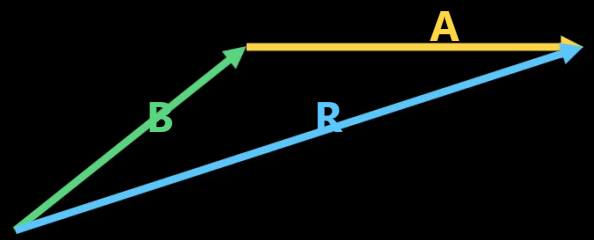
Rough work / extra work

Why drawing order does not change R

A + B and B + A reach the same final point. The path changes, but the start and finish do not.



A then B

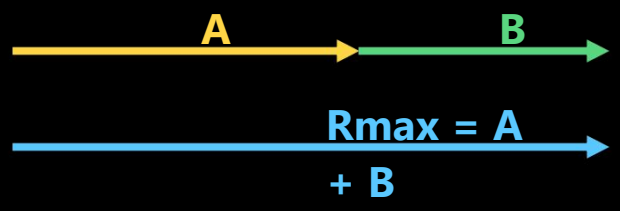


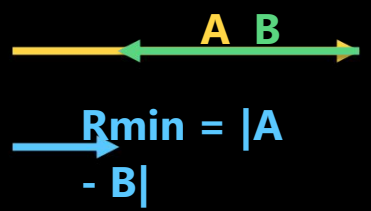
B then A

Same starting point + same ending point = same resultant.

Resultant range: minimum to maximum

The largest resultant happens when both arrows pull in the same direction. The smallest happens when they oppose.





Allowed range

$|A - B| \le R \le A + B$

A resultant outside this range is impossible.

Magnitude of the resultant

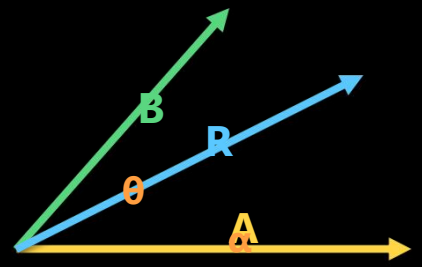
$$R = \sqrt{(A^2 + B^2 + 2\,A\,B\cos\theta)}$$

$$\tan\alpha = (B\sin\theta) / (A + B\cos\theta)$$

α is the angle between A and the resultant.

θ = 0: R max = A + B. θ = 180: R min = |A - B|. θ = 90: R = $\sqrt{(A^2 + B^2)}$.

R is allowed only from |A - B| to A + B. Both ends are included.



Two 4 N forces at 60°

|A| = 4 N, |B| = 4 N, angle = 60°. Find R.

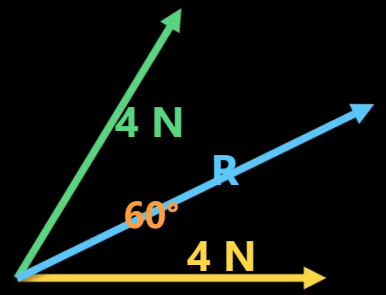
Solution: Two 4 N forces at 60°

Step 1: $R = \sqrt{(16 + 16 + 32 \times \cos 60°)}$

Step 2: $\cos 60° = 1/2$, so the middle term is 16

Step 3: $R = \sqrt{48} = 4\sqrt{3}$

$R = 4\sqrt{3}$ N.



Rough work / extra work

Magnitudes 3 and 4

Two vectors have magnitudes 3 and 4. Find R at 0°, 90° and 180°.

Solution: Magnitudes 3 and 4

Step 1: At 0°, R = 3 + 4 = 7

Step 2: At 90°, R = √(9 + 16) = 5

Step 3: At 180°, R = 4 - 3 = 1

7, 5 and 1.

3 and 4 at 60° and 120°

Same magnitudes 3 and 4. Find R at 60° and at 120°.

Solution: 3 and 4 at 60° and 120°

Step 1: At 60°: $\sqrt{(9 + 16 + 12)} = \sqrt{37}$

Step 2: At 120° cos = -1/2, so $\sqrt{(9 + 16 - 12)} = \sqrt{13}$

$\sqrt{37}$ and $\sqrt{13}$.

5 N east and 10 N north

Two forces, 5 N due east and 10 N due north. Find the resultant.

Solution: 5 N east and 10 N north

Step 1: The angle is 90°, so the cross term is 0

Step 2: $R = \sqrt{(25 + 100)} = \sqrt{125} = \sqrt{(25 \times 5)} = 5\sqrt{5}$

$5\sqrt{5}$ N.

Rough work / extra work

Two 5 N forces at 120°

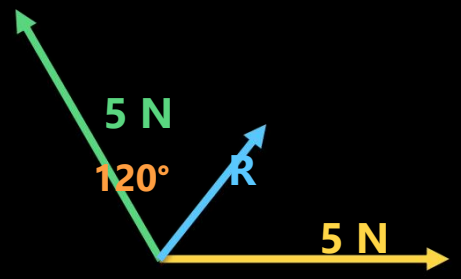
Two forces of 5 N have 120° between them. Find the net force.

Solution: Two 5 N forces at 120°

Step 1: Use the angle between the arrows, which is 120°, not a nearby 60° mark

Step 2: $R = \sqrt{(25 + 25 + 50 \times (-1/2))} = \sqrt{(50 - 25)} = \sqrt{25}$

Net force = 5 N.



Which pair cannot give a resultant of 4?

Pairs: (2, 4), (4, 4), (4, 9), (1, 5). Which cannot give resultant 4?

Solution: Which pair cannot give a resultant of 4?

Step 1: (2, 4) allows 2 to 6. 4 is inside.

Step 2: (4, 4) allows 0 to 8. (1, 5) allows 4 to 6.

Step 3: (4, 9) allows 5 to 13. 4 is smaller than the minimum.

(4, 9) cannot give 4.

Rough work / extra work

Three forces, zero resultant

Which set can have zero resultant: (2, 4, 7), (3, 1, 5), (2, 8, 11), or (3, 4, 2)?

Solution: Three forces, zero resultant

Step 1: The third force must lie between |first - second| and first + second

Step 2: For 3 and 4, the range is 1 to 7. The force 2 lies inside it.

Step 3: 7 is outside 2 to 6. 5 is outside 2 to 4. 11 is outside 6 to 10.

Only (3, 4, 2) can give zero.

Equal magnitudes

If |A| = |B| = A, then R = 2 A cos(θ / 2)

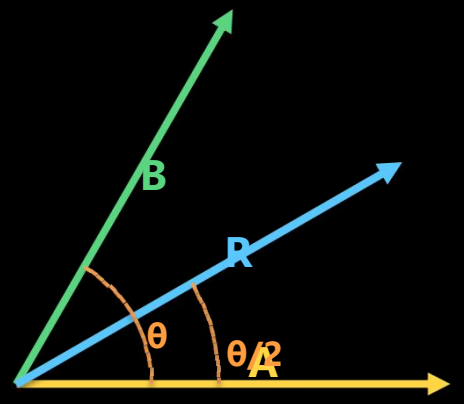
Their difference has magnitude D = 2 A sin(θ / 2)

The resultant then bisects the angle. Angle between A and R is θ / 2.

θ = 0: R = 2A, D = 0. θ = 90: both are √2 A. θ = 120: R = A and D = √3 A. θ = 180: R = 0 and D = 2A.

Equal vectors: resultant bisects the angle

When |A| = |B|, the triangle is symmetric. The resultant must lie exactly in the middle.



$$R = 2A \cos(\theta / 2)$$

Fast memory

Equal vectors make an isosceles triangle, so R is the angle bisector.

Sum and difference have the same magnitude

If |A + B| = |A - B|, what is the angle between A and B?

Solution: Sum and difference have the same magnitude

Step 1: Equal lengths of sum and difference means the cross term must vanish

Step 2: $2AB\cos\theta = 0$, so $\cos\theta = 0$

$90°$.

Rough work / extra work

Two unit vectors add to a unit vector

The sum of two unit vectors is itself a unit vector. Find the magnitude of their difference.

Solution: Two unit vectors add to a unit vector

Step 1: $R = 1$ and each magnitude is 1, so $1 = 2\cos(\theta / 2)$, so $\theta = 120°$

Step 2: $D = 2 \sin 60° = 2 \times \sqrt{3}/2 = \sqrt{3}$

Magnitude of the difference = $\sqrt{3}$.

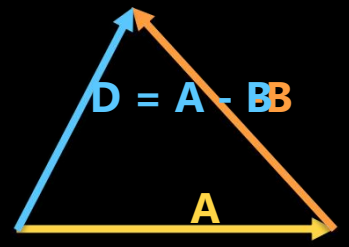
Vector subtraction

A - B = A + (-B). Reverse B, then add.

$$D = \sqrt{(A^2 + B^2 - 2\,A\,B\cos\theta)}$$

$\theta = 0$: $D = |A - B|$, the minimum. $\theta = 180$: $D = A + B$, the maximum. $\theta = 90$: $D = \sqrt{(A^2 + B^2)}$.

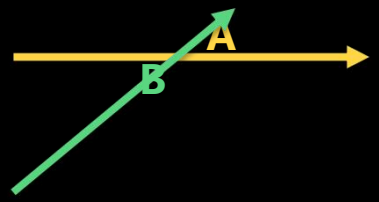
A + B equals B + A. A - B does not equal B - A.



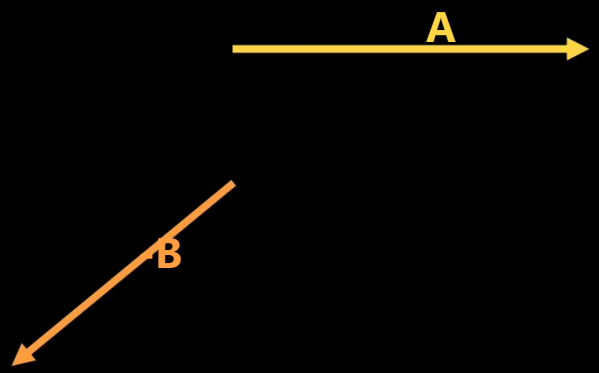
reverse B first

Subtraction is not a new operation

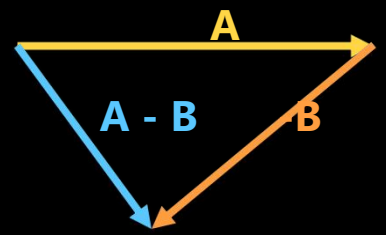
Every subtraction question is an addition question after reversing the second vector.



Step 1: given A and B



Step 2: reverse B



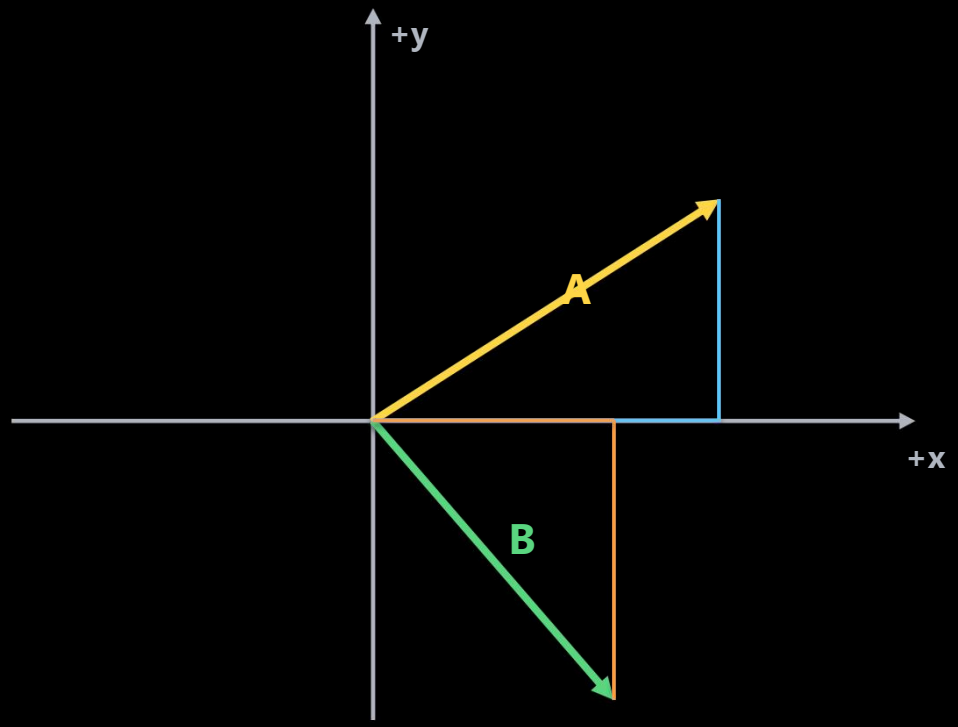
Step 3: add A + (-B)

A - B and B - A have the same length but opposite directions.

Rough work / extra work

Adding by components is adding columns

The x-parts combine with x-parts. The y-parts combine with y-parts. This is the cleanest method for many vectors.



$$\mathbf{A} + \mathbf{B} = (A_x + B_x)\,\hat{\imath} + (A_y + B_y)\,\hat{\jmath}$$

Column idea

x-column decides left-right. y-column decides up-down.

3D same rule

$A + B = (A_x+B_x)\hat{\imath} + (A_y+B_y)\hat{\jmath} + (A_z+B_z)\hat{k}$

Add by components

$A = 4\,\hat{i} + 3\,\hat{j}$ and $B = 3\,\hat{i} - 4\,\hat{j}$. Find $A + B$, $A - B$, and $|A| + |B|$.

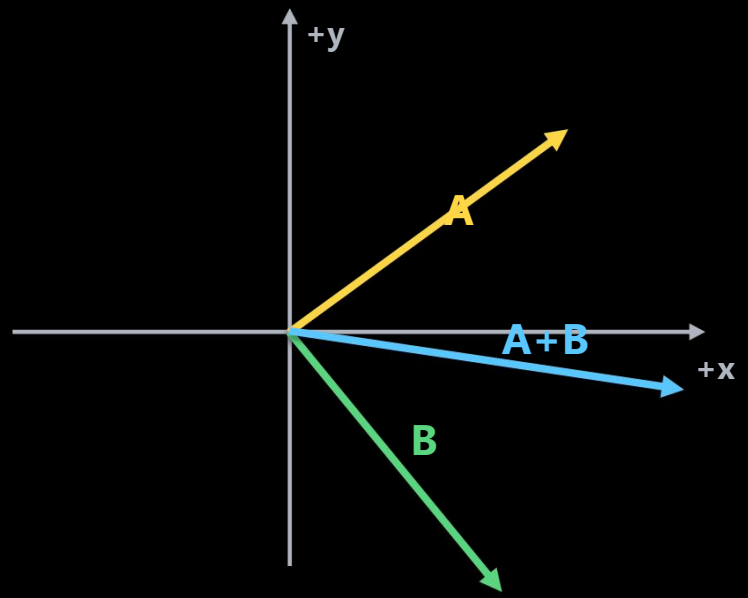
Solution: Add by components

Step 1: $\vec{A} + \vec{B} = 7\,\hat{i} - 1\,\hat{j}$

Step 2: $\vec{A} - \vec{B} = 1\,\hat{i} + 7\,\hat{j}$

Step 3: $|\vec{A}| = 5$ and $|\vec{B}| = 5$, so $|\vec{A}| + |\vec{B}| = 10$

Sum of magnitudes is 10. Magnitude of the sum is √50.



Change in velocity

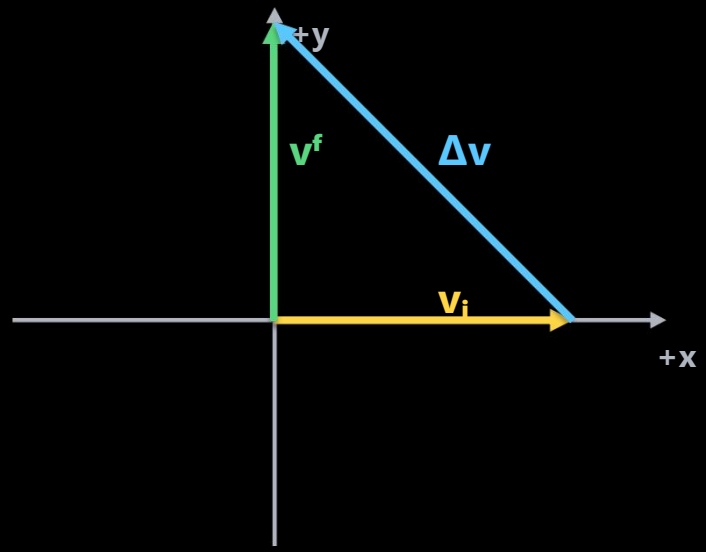
Velocity changes from 10 m/s east to 10 m/s north. Find the change in velocity.

Solution: Change in velocity

Step 1: Change = final - initial = 10 ĵ - 10 î

Step 2: Magnitude = √(100 + 100) = 10 √2

10 √2 m/s, pointing north of west.



Rough work / extra work

Rotate 2 î

A = 2 î is rotated by 90°, 180°, 270° and 360° anticlockwise. Write each new vector.

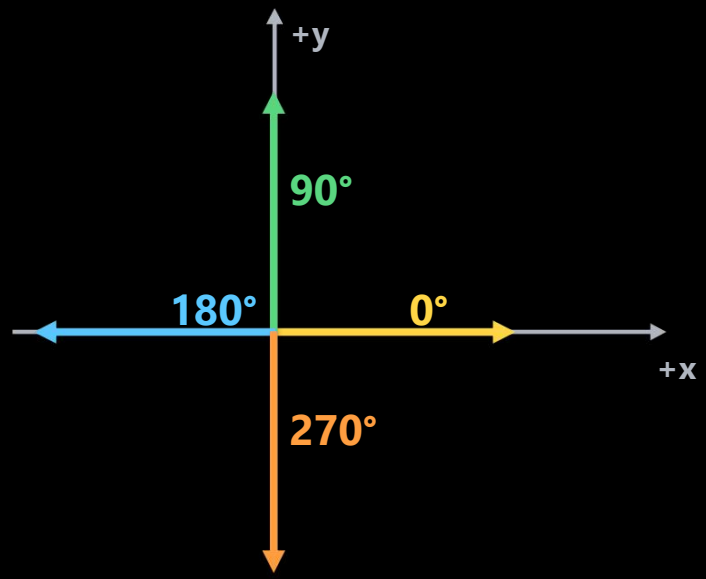
Solution: Rotate 2 î

Step 1: 90°: 2 ĵ

Step 2: 180°: -2 î

Step 3: 270°: -2 ĵ. 360°: 2 î again

Rotation keeps the magnitude and changes only the direction.



Dot product

$$A \cdot B = A\,B \cos\theta = A_x B_x + A_y B_y + A_z\, B_z$$

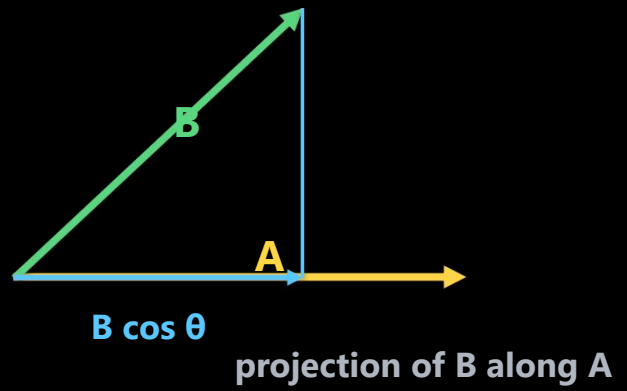
If the dot product is 0, the vectors are perpendicular.

cos θ = (A · B) / (A B). That is how the angle is found from components.

Component of B along A = (A · B) / A.

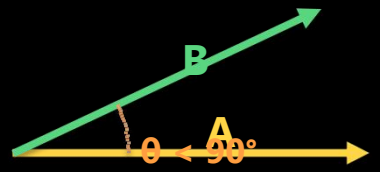
$\hat{i} \cdot \hat{i} = 1$. $\hat{i} \cdot \hat{j} = 0$. The same rule holds for $\hat{j}$ and $\hat{k}$.

A unit vector dotted with itself is 1.

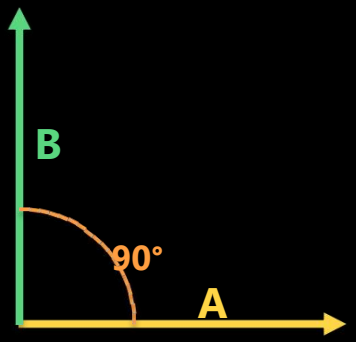


Dot product tells alignment

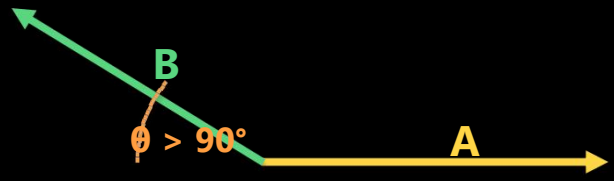
Dot product is positive when vectors face mostly the same way, zero at 90°, and negative when they face opposite way.



$A \cdot B > 0$



$A \cdot B = 0$



$A \cdot B < 0$

Why this matters

It quickly tests perpendicular vectors and helps find the angle between two vectors.

Are these two vectors perpendicular?

A = 3 $\hat{i}$ + 4 $\hat{j}$ + 5 $\hat{k}$ and B = 3 $\hat{i}$ + 4 $\hat{j}$ - 5 $\hat{k}$. What is the angle between them?

Solution: Are these two vectors perpendicular?

Step 1: $A \cdot B = 9 + 16 - 25 = 0$

The angle is 90°.

Rough work / extra work

Angle with 4 $\hat{i}$

$A = 2\hat{i} + 6\hat{j} + 3\hat{k}$ and $B = 4\hat{i}$. Find $\cos\theta$.

Solution: Angle with 4 î

Step 1: $A \cdot B = 8$

Step 2: $|A| = \sqrt{(4 + 36 + 9)} = 7$ and $|B| = 4$

Step 3: $\cos \theta = 8 / 28 = 2/7$

$\theta = \cos^{-1}$ of 2/7.

Unit vectors from components

$\hat{i}$ itself is already a unit vector.

Unit vector of $\hat{i} + \hat{j}$ is $(\hat{i} + \hat{j}) / \sqrt{2}$.

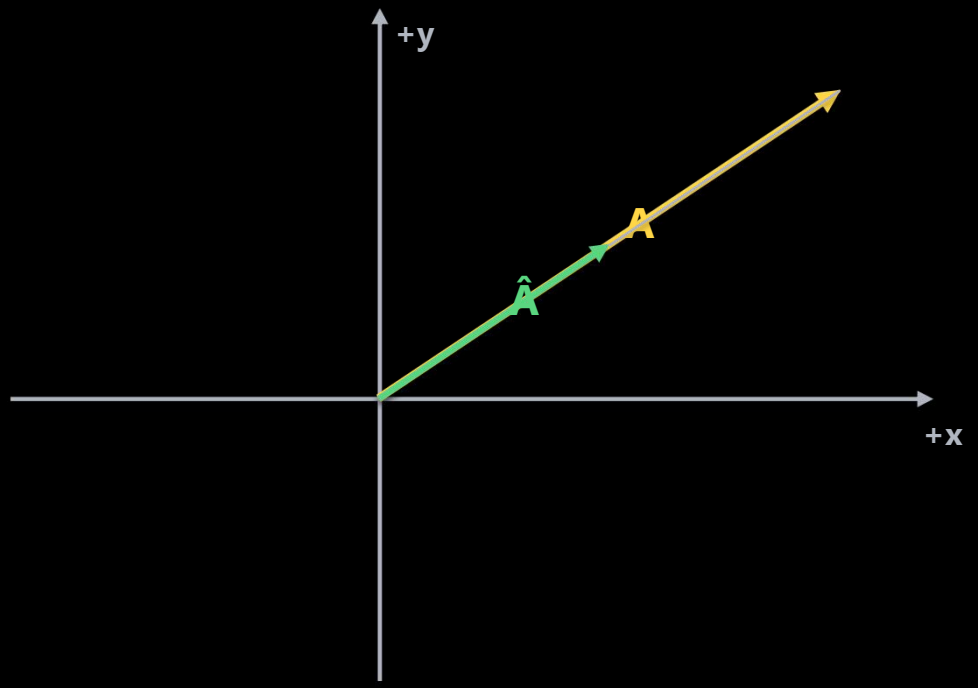
Unit vector of $\hat{i} + \hat{j} - \hat{k}$ is $(\hat{i} + \hat{j} - \hat{k}) / \sqrt{3}$.

$\sin\theta\,\hat{i} - \cos\theta\,\hat{j}$ already has magnitude 1, because $\sin^2 + \cos^2 = 1$.

Unit vector of $3\,\hat{i} + 4\,\hat{j}$ is $(3\,\hat{i} + 4\,\hat{j}) / 5$.

Unit vector keeps direction and removes size

A unit vector is only the direction of a vector. It has length 1 and no physical unit.



$$\hat{A} = A / |A|$$

Example

For $A = 3\hat{i} + 4\hat{j}$, $|A| = 5$, so $\hat{A} = (3\hat{i} + 4\hat{j})/5$.

Do not attach N, m/s or kg to a unit vector.

Rough work / extra work

Perpendicular forces A and A/2

Two forces of magnitudes A and A/2 are perpendicular. Find the magnitude of their resultant.

Solution: Perpendicular forces A and A/2

Step 1: $\cos 90° = 0$, so $R = \sqrt{(A^2 + A^2/4)}$

Step 2: $R = \sqrt{(5\,A^2 / 4)} = (A/2)\,\sqrt{5}$

$(\sqrt{5} / 2) \times A$.

Resultant is half the larger vector

The resultant of two vectors is half the larger one, and it is perpendicular to the smaller one. Find the angle between the vectors.

Solution: Resultant is half the larger vector

Step 1: R perpendicular to the smaller vector means $(A + B) \cdot B = 0$

Step 2: $\cos\theta = -B/A$

Step 3: $R = A/2$ then gives $B = (\sqrt{3}/2)\,A$

$\cos\theta = -\sqrt{3}/2$, so $\theta = 150°$.

Acceleration of a 4 kg body

$F_1 = 5\,\hat{i} + 8\,\hat{j} + 7\,\hat{k}$ and $F_2 = 3\,\hat{i} - 4\,\hat{j} - 3\,\hat{k}$ act on 4 kg. Find the acceleration.

Solution: Acceleration of a 4 kg body

Step 1: Net force = 8 $\hat{i}$ + 4 $\hat{j}$ + 4 $\hat{k}$

Step 2: a = net force / 4

a = 2 $\hat{i}$ + 1 $\hat{j}$ + 1 $\hat{k}$.

Rough work / extra work

Vector 02 lock lines

Components: cos along the axis you measured from, then keep the sign

$|F| = \sqrt{(F_x^2 + F_y^2 + F_z^2)}$

$R = \sqrt{(A^2 + B^2 + 2AB \cos \theta)}$, and $|A-B|$ to $A+B$ is the allowed range

$A \cdot B = 0$ means 90°. Unit vector = vector divided by magnitude.

Practice

1. Draw 30° east of south and 60° south of east on one compass.

2. Resolve 10 N at 30° and at 53°, with signs.

3. Redo magnitudes 3 and 4 at 0°, 60°, 90°, 120° and 180°.

4. Redo the unit-vector list and the 4 kg acceleration.

Do not add magnitudes when the arrows are not along one line.